



Knee Osteoarthritis Prediction and Progression using Multi-Modal Deep Learning.

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Our Team

25 - 26J - 112

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**THIS PROJECT INTRODUCES
AN INTELLIGENT SYSTEM FOR**

PROBLEM STATEMENT & STAKEHOLDERS

OBSERVED ISSUES

Current KOA diagnosis relies mainly on **X-ray, MRI, and clinical examination**

WHY IT MATTERS?

- Expensive diagnostic methods
- Limited accessibility in rural areas
- No continuous monitoring of disease progression
- Delayed treatment decisions

STAKEHOLDERS

- Primary users
 - Orthopedic doctors
 - Patients
- Secondary users
 - Medical researchers
 - Healthcare providers
- Decision makers
 - Hospitals
 - Healthcare organizations

RESEARCH GAP & OBJECTIVES

- Existing KOA prediction systems rely on a single data modality
 - **X-ray images**
 - **MRI scans**
 - **Clinical data**
- In many cases, disease diagnosis also depends heavily on the doctor's knowledge and clinical experience, which may lead to variability in decisions.

Main Objective

Develop a multi-modal AI-based system that integrates

- Medical imaging (X-ray & MRI)
- Clinical and biomarker data
- Wearable IoT sensor signals



To improve

- KOA detection accuracy
- Severity grading
- Continuous patient monitoring

SOLUTION OVERVIEW

Solution Summary

An AI-driven multimodal system for Knee Osteoarthritis prediction, severity grading, and monitoring.

Key Features Delivered

FR1 — Clinical Data Prediction

- Machine learning model using demographic, clinical and biomarker data

FR2 — Imaging Analysis

- KOA detection using X-ray and MRI deep learning models

FR3 — Severity Grading

- KOA severity classification using Kellgren-Lawrence grading system

FR4 — IoT Monitoring

- Wearable system using VAG signals for real-time knee monitoring

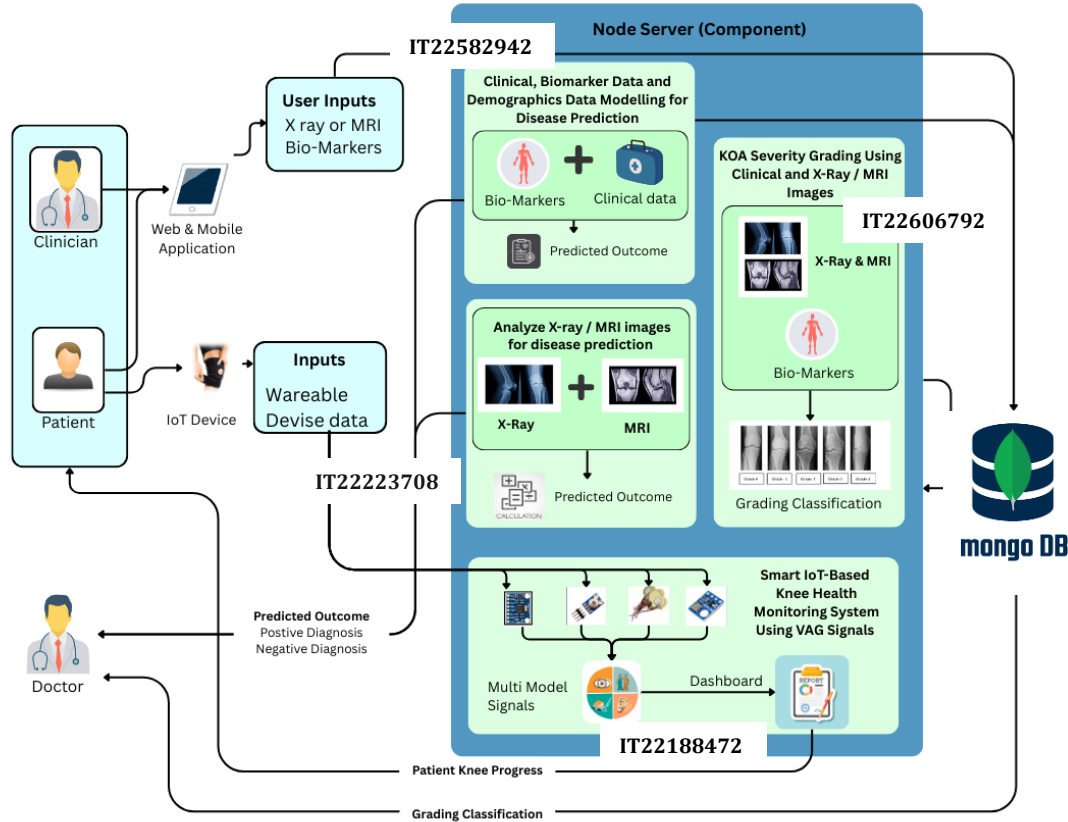
Non-Functional Requirements

- Reliability & Robustness
- Performance & clinical utility
- Regulatory compliance

Current Completion Status

Approximately **85–90% completed**
Components developed and integrated.

SYSTEM ARCHITECTURE & INTEGRATION



APIs

- REST APIs

Data Formats

- JSON
- Multipart Form Data
- NumPy Arrays

COMMERCIALIZATION / IMPACT

Target Users

- Hospitals
- Orthopedic clinics
- Physiotherapy centers
- Healthcare researchers

Value Proposition

- Early KOA detection
- Accurate severity assessment
- Continuous monitoring outside hospital environments
- Low-cost AI healthcare solution

Cost & Feasibility

Low-cost implementation using

- Open-source AI frameworks
- Affordable IoT sensors
- Cloud or local deployment

Adoption Plan

- Clinical pilot testing
- Integration with hospital systems
- Deployment in healthcare environments

Thanks

